# Module 3: Critical Thinking

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## Critical Thinking Assignment (75 Points)

## Case Studies of Large Data Files

In the **Portfolio Project (https://csuglobal.instructure.com/courses/123286/assignments/2269095)** , we are working with data files containing random numbers. The data files can be anywhere from a few lines (rows) to millions or billions of rows. Creating and processing such files efficiently has very practical applications. For example, instead of random numbers, a company might have names and other information for its employees, which can be in the hundreds of thousands.

Describe, in detail, 3 real-world examples of possible data files of at least one million rows, and why it would be important for a business to be able to process such files in fractions of a second instead of several seconds. You can use arenas such as financial systems, inventory tracking, state drivers license information, or anything else that you can think of. How would you use the methods that were covered in this and the previous module to optimize processing times for such files?

### Grading Criteria

- Your paper should be 2-3 pages in length, not including the cover page and references page.
- Your paper must be formatted according to APA guidelines in the CSU Global Writing Center (available in the left-hand navigation panel).
- Your claims should be supported by evidence. Include 2-3 credible references. The CSU Global Library (available in the left-hand navigation panel) is a good place to find these references.
- All references must be cited in the text and listed on the references page, according to APA formatting.
- See the rubric below for more details about how you will be graded for this assignment.

**CSC507_M03_CT**

| Criteria | Ratings | | | | Pts |
|---|---|---|---|---|---|
| Real-World Examples of Large Data Files | **20 pts**<br>**Meets Expectation**<br>Provides three detailed, realistic examples (each ≥ 1 million rows) that clearly demonstrate real-world applications (e.g., financial, logistics, or state systems). Each example includes thoughtful reasoning on data scale and structure. | **16 pts**<br>**Approaches Expectation**<br>Provides two or three examples, but with limited detail or weaker connection to real-world applications. Some examples may lack clear explanation of scale or relevance. | **12 pts**<br>**Below Expectation**<br>Provides one or two vague examples with minimal real-world connection or unclear data scale. | **8 pts**<br>**Limited Evidence**<br>Provides few or no examples; examples are irrelevant or missing entirely. | 20 pts |
| Explanation of Business Importance (Speed & Efficiency) | **15 pts**<br>**Meets Expectation**<br>Clearly and convincingly explains why processing large files quickly is critical in each scenario. Demonstrates strong understanding of real-time needs, efficiency, and business impact. | **12 pts**<br>**Approaches Expectation**<br>Provides some explanation of importance, but reasoning is general or lacks depth for one or more examples. | **9 pts**<br>**Below Expectation**<br>Mentions importance superficially, with limited or inaccurate understanding of efficiency and performance impacts. | **6 pts**<br>**Limited Evidence**<br>Shows little or no understanding of why processing speed matters. | 15 pts |
| Application of Course Methods for Optimization | **20 pts**<br>**Meets Expectation**<br>Thoroughly discusses specific methods or techniques (from current and prior | **16 pts**<br>**Approaches Expectation**<br>Identifies relevant methods, but explanation of how they optimize performance is | **12 pts**<br>**Below Expectation**<br>Mentions methods but shows confusion or inaccuracy about | **8 pts**<br>**Limited Evidence**<br>Provides no meaningful discussion of optimization methods. | 20 pts |

| Criteria | Ratings | | | | Pts |
|---|---|---|---|---|---|
| | modules) for optimizing file processing. Shows clear understanding of how these improve performance. | limited or partially correct. | their purpose or implementation. | | |
| Use of Sources & APA Formatting | **10 pts**<br>**Meets Expectation**<br>Incorporates 2–3 credible sources. Sources are well-integrated and cited accurately in APA format (both in-text and references). | **8 pts**<br>**Approaches Expectation**<br>Uses required sources, but citations or references contain minor errors or sources are weakly integrated. | **6 pts**<br>**Below Expectation**<br>Uses fewer than 2 sources or sources of questionable credibility; multiple APA format errors. | **4 pts**<br>**Limited Evidence**<br>No credible sources used or APA format not followed. | 10 pts |
| Organization, Clarity, & Mechanics | **10 pts**<br>**Meets Expectation**<br>Paper is 2–3 pages, well-organized, and clearly written. Logical flow, strong transitions, and few or no grammatical errors. | **8 pts**<br>**Approaches Expectation**<br>Generally organized and clear, though may have minor issues with flow or clarity. Some grammatical or spelling errors. | **6 pts**<br>**Below Expectation**<br>Organization is weak or uneven. Writing contains frequent errors that distract from meaning. | **4 pts**<br>**Limited Evidence**<br>Disorganized or incomplete; major grammatical or structural problems make content difficult to follow. | 10 pts |
| | | | | | Total Points: 75 |